

Recitation 8 - Simulating Jump Processes (pt 2) - Amer Al-Hiyar

Recap

$\Rightarrow$ A Markov jump process is a sequence of states $\{x_i\} \in \mathcal{S}$ and a sequence of dwell times

$\{\tau_i\} \in \mathbb{R}^+$. The state space $\mathcal{S} = \{\phi_1, \phi_2, \dots, \phi_n\}$ is finite or countably infinite.

$\Rightarrow$ If the system is in state i , the dwell time is exponentially distributed according to the escape rate

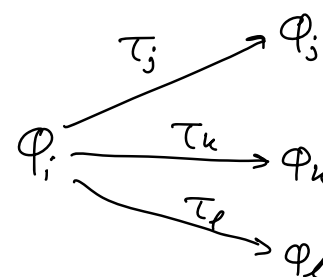
$$\tau \sim \text{Exp}(r(\phi_i)), \quad r(\phi_i) = \sum_{j \neq i} w(\phi_i \rightarrow \phi_j)$$

The next state is chosen independently of τ according to:

$$P_{i \rightarrow j} = \frac{w(\phi_i \rightarrow \phi_j)}{r(\phi_i)}$$

$\Rightarrow$ Equivalent characterization: For each possible transition from ϕ_i to another state ϕ_j , draw a putative reaction time

$$\tau_j \sim \text{Exp}(w(\phi_i \rightarrow \phi_j))$$



The dwell time and next step are chosen according to:

$$\tau = \min \{\tau_j\}$$

$$\text{next state} = \arg \min_j \{\tau_j\}.$$

To prove equivalence, we must show

Consider a collection of n independent exponentially-distributed random variables with rates $k_1, k_2, \dots, k_n$; that is, $\{\tau_1, \dots, \tau_n\}$ with $\tau_i \sim \text{Exp}(k_i)$

(1) The minimum $\tau \equiv \min \{\tau_i\}$ is $\text{Exp}(r)$ distributed with $r = \sum_i k_i$.

(2) The probability that τ_i is the smallest among the $\{\tau_j\}$ is:

$$\mathbb{P}[\arg \min_j \{\tau_j\} = i] = k_i / r$$

Equivalence of next reaction methods

For (1), it suffices to prove the statement for $n=2$, because it may then be applied

pairwise $\min(\tau_1, \tau_2, \dots, \tau_n) = \min(\min(\tau_1, \tau_2), \tau_3, \dots, \tau_n) = \min(\min(\min(\tau_1, \tau_2), \tau_3), \tau_4, \dots, \tau_n)$

and so on. For $n=2$, we have

$$\begin{aligned} \mathbb{P}(\min(\tau_1, \tau_2) > t) &= \mathbb{P}(\tau_1 > t) \mathbb{P}(\tau_2 > t) = \int_t^\infty ds k_1 e^{-k_1 s} \int_t^\infty ds' k_2 e^{-k_2 s'} \\ &= e^{-k_1 t - k_2 t} \end{aligned}$$

$$= 1 - \mathbb{P}(\min(\tau_1, \tau_2) < t)$$

we thus identify the CDF as that of an $\text{Exp}(k_1 + k_2)$ R.V. $\square$

For (2), we use indicator variables

$$\mathbb{P}[\arg\min_j \{\tau_j\} = i] = \left\langle \prod_{p \neq i} \mathbb{1}(\tau_p > \tau_i) \right\rangle$$

where $\mathbb{1}(A) \equiv 1$ if event A occurs and zero otherwise.

The terms within the product are not independent: if I tell you $\tau_k > \tau_i$ for some k , this tells you something about the smallness of τ_i , so that $(\tau_m > \tau_i)$ is now more likely.

To resolve this, we condition on τ_i and use the tower rule $\mathbb{E}(X) = \mathbb{E}_Y[\mathbb{E}_X[X|Y]]$:

$$\begin{aligned} \mathbb{P}[\arg\min_j \{\tau_j\} = i] &= \left\langle \left\langle \prod_{p \neq i} \mathbb{1}(\tau_p > \tau_i) \mid \tau_i \right\rangle \right\rangle_{\tau_i} \\ &= \left\langle \prod_{p \neq i} \mathbb{P}(\tau_p > \tau_i) \right\rangle_{\tau_i} \\ &= \left\langle \prod_{p \neq i} e^{-k_p \tau_i} \right\rangle_{\tau_i} \\ &= k_i \int_0^\infty dt \exp\left(-\sum_{j=1}^n k_j t\right) \\ &= \frac{k_i}{\sum_{j=1}^n k_j} \end{aligned}$$

$\square$

2.3.1: "First reaction" method

Based on the above Gillespie proposed the following algorithm: at each step draw an $\text{Exp}(w(x \rightarrow \varphi))$ variable τ_φ for every $\varphi \in \mathcal{S}$. Set $t \rightarrow t + \min \{\tau_\varphi\}$ and $x \rightarrow \arg\min_\varphi \{\tau_\varphi\}$.

This algorithm is obviously less efficient because it draws up to $|\mathcal{S}|$ random variables at each step and identifies their minimum. Its usefulness is thus mostly pedagogical.

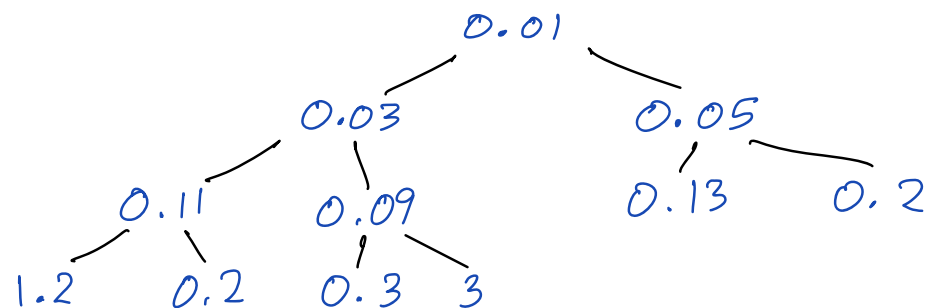
2.3.2: "Next reaction" method (Gibson and Bruck, 2000).

The idea here is to draw all the possible transition times $\{\tau_i\}$ and to store them in an efficient data structure that maintains their order.

At each step, the smallest τ_i transition is executed, and only those reactions which are affected are redrawn.

The original implementation is specialized to chemical reactions, so we list here only the main idea:

⇒ The transition times are stored in a *min heap* data structure (usually indexed)



⇒ The next reaction is always the root and can be found in $\mathcal{O}(1)$ time.

⇒ An additional data structure keeps track of the dependency structure between the elements (e.g. nearest neighbor sites on a lattice).

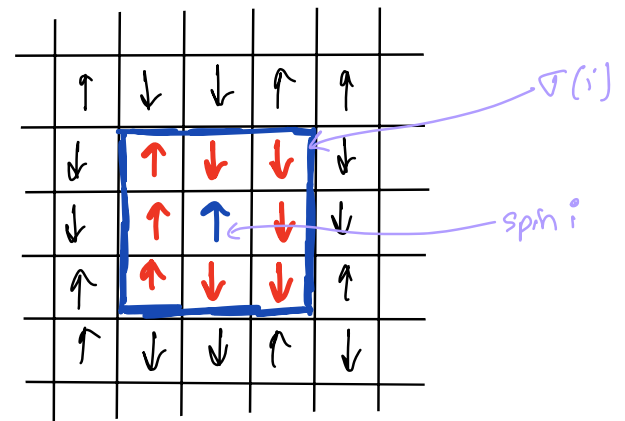
Example: spins on a lattice (e.g. a kinetic Ising model):

Consider a model with spins $s = \pm 1$ on a lattice $\mathbb{Z}^d$.

Spin i flips at a rate which is a function of s_i

and the nearest neighbors of i , denoted $\sigma(i)$:

$$\text{Flip rate} = w(\{s_j : j \in \sigma(i)\}) \equiv w_i(\vec{s})$$



This can be simulated as follows:

1) Initialize the spin lattice. For each $i \in L^d$, draw the flip

time of spin i according to $\tau_i \sim \text{Exp}(w_i(\vec{s}))$

2) Organize the L^d pairs (i, τ_i) into a min heap sorted according

to τ_i . Create a data structure of pointers to the tree for each i .

3) Set the simulation time to $t = \tau_j$, where $j = \text{argmin}_i \{\tau_i\}$.

4) Flip spin j .

5) For each $i \in \sigma(j)$ and for j itself, redraw τ_i as $\tau_i \rightarrow t + \Delta\tau_i$, where

$\Delta\tau_i \sim \text{Exp}(w_i(\vec{s}))$. Re-organize the heap. Return to step 3.

The slowest step is the time to reorganize the heap, which is $\mathcal{O}(\log N)$ instead of $\mathcal{O}(N)$.

This is a big improvement over the direct method

2.4 Discrete-time (rejection) algorithms

In cases where the number of possible reactions is very large $n \gg 1$ and all the reactions have comparable rates, it is often most efficient to use a discrete-time algorithm which converges to the jump process as $\Delta t \rightarrow 0$.

We will focus on many-body lattice models like the kinetic Ising model. For such models, a useful discrete-time algorithm is **random sequential update**, where particles/spins are randomly chosen and updated according to some probabilistic rule.

For a kinetic Ising model with flip rates $w_i(\vec{s})$ at site i , the algorithm is as follows: Pick a time step Δt . Then

- 1) Advance time $t \rightarrow t + \Delta t/N$, where $N \equiv L^d$
- 2) Randomly select a lattice site i (**proposal**)
- 3) With probability $w_i(\vec{s})\Delta t$, flip spin i . (**acceptance**)
[with probability $1 - w_i(\vec{s})\Delta t$, do nothing. (**rejection**)]

Return to (1).

How do we choose Δt ? We would like for it to be as large as possible while keeping the acceptance/rejection probabilities $\in [0, 1]$. Thus $\Delta t = 1/w_{\max}$, where $w_{\max}$ is the maximum possible value of w_i .

The above describes a discrete-time **Markov Chain** that approximates the jump process as $N \rightarrow \infty$ (or as $\Delta t \rightarrow 0$). To see this, let C denote a configuration of spins, and C_i that same configuration with spin i flipped. Then the probability of a configuration obeys the following recursion relation in discrete time:

$$P(C, t + \frac{\Delta t}{N}) = \sum_{i=1}^N P(C_i, t) \frac{w_i(C_i) \Delta t}{N} + P(C, t) \left(1 - \sum_{i=1}^N \frac{w_i(C) \Delta t}{N}\right)$$

The continuous time limit follows by expanding $P(C, t + \frac{\Delta t}{N}) = P(C, t) + \partial_t P(C, t) \frac{\Delta t}{N}$, valid when $\Delta t/N$ is $\ll$ the timescale of variations in P (e.g. for $\frac{\Delta t}{N} \ll 1/w_{\max}$).

We obtain:

$$\begin{aligned}
\partial_t P(c, t) &= \sum_{i=1}^N p(c_i, t) w_i(c_i) - p(c, t) w_i(c) \\
&= \sum_{c' \neq c} p(c', t) w(c' \rightarrow c) - p(c, t) w(c \rightarrow c'),
\end{aligned}$$

where we have used that the only nonzero $w(c \rightarrow c')$ are those for which $c' = c_i$ for some i (all transitions involve single flips of some spin). We have thus recovered the master equation.